

Executive Summary

Everyone now lives inside the same paradox: reachable at every second of the day, yet quietly convinced no one is actually there. The WHO links social disconnection to roughly 100 deaths every hour worldwide, and 72% of Filipino youth report loneliness despite averaging 7.5 hours a day online — the more we stay online for each other, the less we actually reach each other. **KIBO is a mobile app where users co-own a living virtual aquarium. It features an on-device AI listener that reads only interaction patterns (never content, never location) and gently cues each person in the other's love language — through wordless memos, mood-driven tank states, and a Connect Moment that rewards putting the phone down when they are finally in the same room.** Where today's tools deliver more forced togetherness, KIBO delivers the opposite — streamlined connection — returning technology to what it should've been all along: **a bridge to real life, not a substitute for it.**

The Problem: Ironically Disconnected in a Constantly-Connected World

We have never been more reachable, yet a WHO analysis finds **1 in 6 people worldwide** now feels lonely — a burden linked to roughly **100 deaths every hour** [1]. In the Philippines, the 2026 Mind Health Report found **72% of Filipino youth** feel lonely and sad despite averaging **7.5 hours of daily screen time** [2]. When we surveyed peers, the paradox showed up live — people said they were happy with tools like Messenger (4.06/5), but they still hold back from reaching out because they don't want to bother busy loved ones (3.33/5 overall, 3.50 among parent-child pairs). The pattern is clear across cultures: connectivity is up, connection is down. We frame this as one main problem with three feeding it.

Sub-problem 1 — Quality time is squeezed by the obligation to always be online

Even when we are physically together, phones sit between us. Field experiments show that when phones are simply **present at the table** — face-down, untouched — people report the meal as less enjoyable, less connected, and lower in empathy than identical meals without phones [3]. **40% of Americans** say they are bothered by how much time their partner spends on the phone; phubbed partners report lower satisfaction, more conflict, and greater emotional distance [4]. In-person time still happens; its quality has quietly collapsed.

Sub-problem 2 — Convenience has made true connection feel too tiring

A generation raised on frictionless everything now finds relational effort disproportionate. Research on friendship formation identifies **convenience** as the single most important determinant of interaction frequency — and emotional depth as the thing convenience specifically avoids [5]. The result is a *friendship recession*: **17% of Americans now report zero close friends**, up from 1% in 1990 [6]. When starting a conversation, driving to see someone, or composing a heartfelt message all feel like too much, relationships default to the lowest-effort medium — and stall there.

Sub-problem 3 — Digital-first communication has eroded interpersonal habits

Heavy online communication correlates with lower empathy in face-to-face settings; teens with high screen time show measurable deficits in reading facial expressions and tone [7]. **65% of Gen Z** say they had to *relearn* basic social skills after the pandemic [8]. Text lets us edit, delay, delete — real conversations have no backspace button, so they start to feel scarier than they used to. The habit muscles for warmth, repair, and small gestures have atrophied precisely when we need them most.

Why existing tools do not fix this

Existing tool	What it offers	Why it misses the disconnection problem
Life360 / GPS trackers	Live location for family circle	Feels like surveillance; erodes autonomy; solves logistics, not closeness

Existing tool	What it offers	Why it misses the disconnection problem
Messenger / social apps	Instant messaging on demand	Threads collapse into logistics ("did u eat?"); heavier topics get avoided until threads go dead
Scheduled family / friend time	Reserved days for bonding	High time cost, forced vulnerability; connection "for the sake of it" rather than genuine
Habit / self-care apps (Finch, Duolingo)	Ambient gamified maintenance of one habit	Single-player: the relationship itself is never the shared object being kept alive

The gap: there is no low-effort, high-privacy channel where people who care about each other can feel each other's presence in minutes-per-week — and be gently pulled back into shared physical time. That is the gap KIBO fills.

Core Concept: The Relationship as a Living Thing You Keep Alive Together

KIBO's central object is a **shared aquarium** rendered identically on every member's phone. It supports pairs by default and small groups (families of 3–4, close friend circles) as the same architecture. The shared aquarium idea resonated in our survey — 4.06/5 across the board, jumping to 4.67 among romantic partners. And the features that ask the least of users scored the highest: a one-tap Send Warmth (4.26/5) and no-reply memos (4.06/5, up to 4.67 for couples). People want to feel loved without having to write anything back.

The design rests on the **Four Stages of Habit** (cue → craving → response → reward) — so maintaining the relationship becomes habitual rather than obligatory — and on **calm technology** as an ambient mediator that informs without demanding attention [9]. IDC projects **60% of households in developed nations** will feature ambient tech by 2026 [10]; KIBO applies that future to the oldest network there is: the people we love.

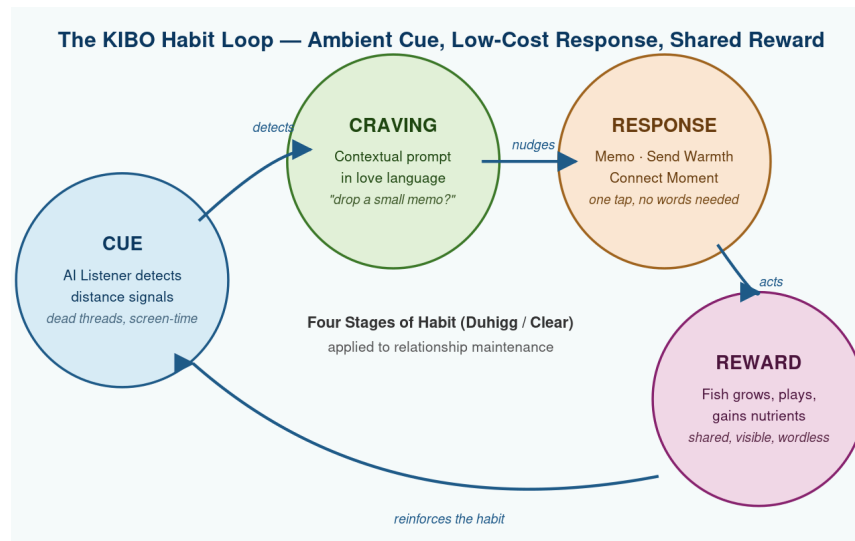


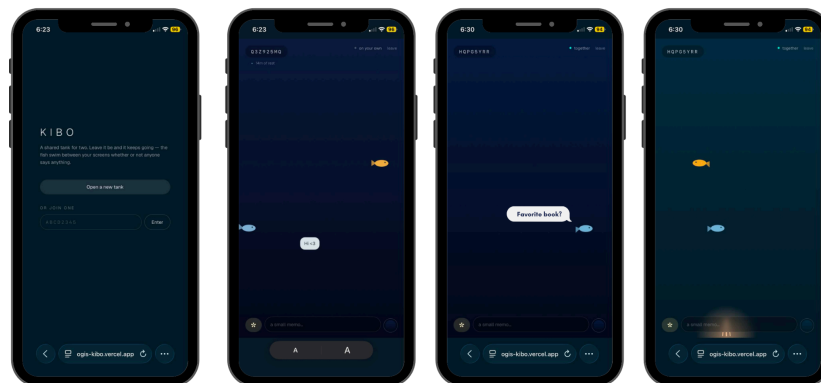
Figure 1. The KIBO habit loop — each feature is one stage of a self-repeating cycle.

Key features

- **AI Listener (cue engine).** Detects emotional-distance signals — dead in-app threads, screen-time spikes — and sends contextualized, love-language-aware prompts: “Leo, a college student, has a tough week ahead. Drop a small memo?” All raw signals are processed on-device; only abstract cues leave the phone.
- **Connect Moment.** A dedicated bonding session that incentivizes real synchronous presence. Triggered two ways: *proximity ping* — when two or more members are physically nearby, both phones surface a matching prompt card; or *user code* — either person enters the other's short code to invite them into a bonding moment across any distance.

- **Connecting — fishes play and gain nutrients.** During a Connect Moment the shared aquarium comes to life: the fishes swim together, play, and recover nutrients they lose over quiet stretches. The reward is visible, shared, and non-verbal.
- **Continuity commitment.** Once both agree to connect, KIBO gently notes that leaving the app will stop the fishes from recovering — then offers a *phone-off* option that keeps the session running quietly in the background, so users can be fully present with each other in the physical world without penalty.
- **Contextual in-moment prompts.** While Connect Mode is active, the fish surfaces light prompts based on what the pair are doing right then — a shared meal, a walk, a study session — turning ordinary time together into small rituals rather than another silent scroll.
- **Small memos.** One-way, no-reply-needed affirmations that arrive as floating bubbles near the fish. The receiver can answer with a single-tap minimized response (a heart, a warm emoji) — zero pressure to compose text.
- **Emotional weather / tank states.** The user signals mood by changing the water — “Deep Sea Deep Blue” for overwhelmed days. The other person sees the color shift in realtime and can respond with a wordless *Send Warmth* tap that drops glowing coral into the tank. Feeling seen, without a single awkward message.
- **Privacy-preserving activity sharing.** A solo hike registers as a milestone — “hike completed” — never as GPS coordinates. Independence and reassurance stop being opposites.
- **Time capsules.** Short-cycle capsules unlock immediately after shared or parallel activities (a photo of the horizon from the teen's hike), converting solo moments into shared memories.
- **Fish points & real-world vouchers.** Interaction quality — reciprocity, not raw volume — grows the fish and earns points redeemable for local café vouchers (Phase 2 partnerships): digital connection deliberately cashing out into physical reconnection.

Ambient Signals, One-Tap Responses, Shared Presence



Concept Screens Based on Real Prototype Screenshots

Used device: iPhone 11

Figure 2. Concept screens — mood signal, one-tap warmth response, and a Connect Moment triggered by proximity ping.

A day in the loop

Thursday, 9 PM. The AI listener notices Leo's threads have gone quiet during exam week and prompts his mother (words-of-affirmation is Leo's love language): she drops a memo — “*Thinking of you today, no need to reply.*” A bubble glows in Leo's tank; he taps once, a heart floats back, the fish eats the bubble and grows. **Saturday.** Leo hikes alone — his independence intact, his mother reassured by a “hike completed” milestone, never a map pin. **Sunday night.** Overwhelmed, Leo dims the tank; his mother sends warmth. **Following Wednesday.** Leo and his mom cross paths at home for the first time all week — both phones ping a Connect Moment. They agree, tap *phone-off*, and the fishes play quietly in the background while they actually eat dinner together. Three connections, one shared meal, zero conversations demanded — one relationship quietly maintained.

Practical Applications & Real-World Value

- **Households under smartphone pressure:** teens and parents who share space but not attention; Connect Moments turn "same room, different phones" into shared presence.
- **Long-distance dyads (partners, close friends, OFW families):** presence across time zones without scheduling video calls; the tank is always on in both hemispheres — a natural fit for the Philippines' 2.19M overseas workers [11].
- **Transition moments:** a child moving out for university, friends graduating into different cities, a partner starting shift work — the highest-risk windows for relationships to fall into logistics-only contact.

The concept also scales as an idea: KIBO argues the future of communication technology is not more bandwidth but **better silence** — ambient, calm-technology channels for any relationship where people care deeply but talk badly. It aligns with the theme of software as a mediator of human wellbeing, not just information flow.

Viability: The Path to Profitability

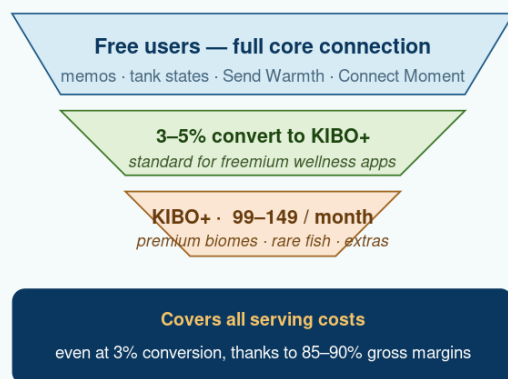
The commercial model is proven adjacently. **Finch**, a virtual-pet self-care app built on the same habit-loop psychology, reached roughly **US\$30M in annual recurring revenue** without venture capital on a simple freemium model (US\$9.99/month premium tier) [12] — while serving only individuals. KIBO applies the same mechanics to relationships *between* people, where the emotional switching cost of abandoning a fish you raised together makes retention structural rather than incidental. Parents and kids especially want reassurance without being watched. Our survey shows sharing moments like "hike completed" milestones instead of GPS pins scored 4.50/5 among parent–child pairs (who separately flagged tracking apps as a real pain, 3.60/5). And the single most-loved feature across everyone? Earning points for real café dates together — 4.29/5 overall, 4.67/5 among couples.

Freemium model. Core connection features — memos, tank states, Send Warmth, Connect Moment, activity milestones — stay free forever; we never paywall the relationship itself, only its decoration. **KIBO+** (₱99–149/month) unlocks premium tank biomes, rare fish species, memo styles, extended time-capsule storage, and expanded small-group support beyond 2 members. **Unit economics** are software-typical: Firebase's free tier absorbs pilot usage; at scale, infrastructure plus AI nudges (Claude Haiku, short and infrequent) cost only a few pesos per member per month, with a rule-based fallback throttling AI spend without losing core function. Gross margins of 85–90%+ mean a conversion rate of just 3–5% of active members — standard for freemium wellness apps — covers all serving costs. Incentives stay clean: we are paid by subscriptions, never by attention or ads — profit and purpose point the same direction.

Go-to-market. Phase 1 (pilot): 5–10 dyads and small groups via our university community, validating retention and relational outcomes. Phase 2 (local launch): campus organizations and family associations, with 2–3 partner cafés proving the voucher loop as a growth flywheel. Phase 3 (scale): direct-to-consumer growth in urban PH markets, then Southeast Asian regional expansion using the same freemium engine.

Freemium Unit Economics — Small Conversion, Structural Retention

The Conversion Funnel



Core connection features stay free forever — we never paywall the relationship.

Per-User Monthly Unit Economics

REVENUE (KIBO+ member)

99 – 149

100%

SERVING COST (infra + AI nudges)

10–15

~10–15%

GROSS MARGIN

85 – 90%

margin

Precedent: Finch — ~US\$30M ARR, bootstrapped
same habit-loop psychology, single-player. KIBO applies it to two.

Feasibility: Buildable by Students, This Semester

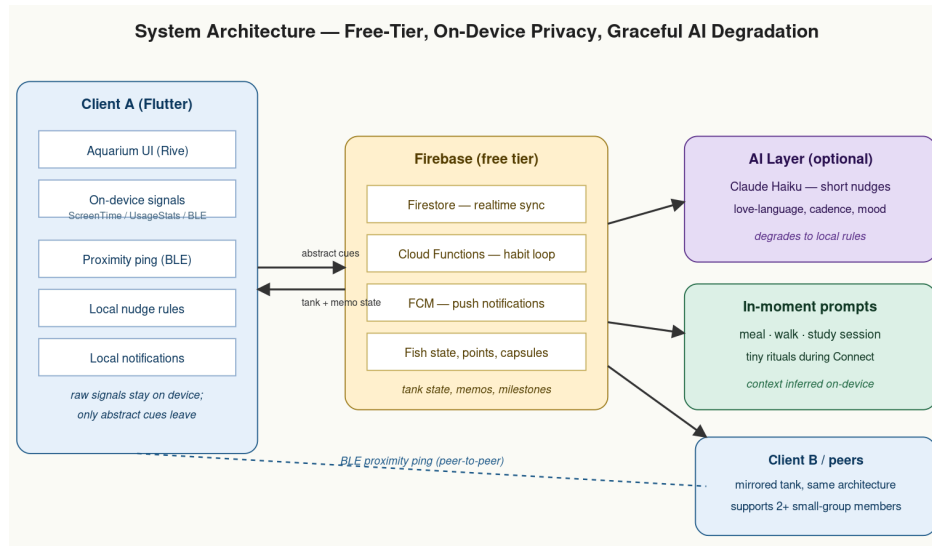
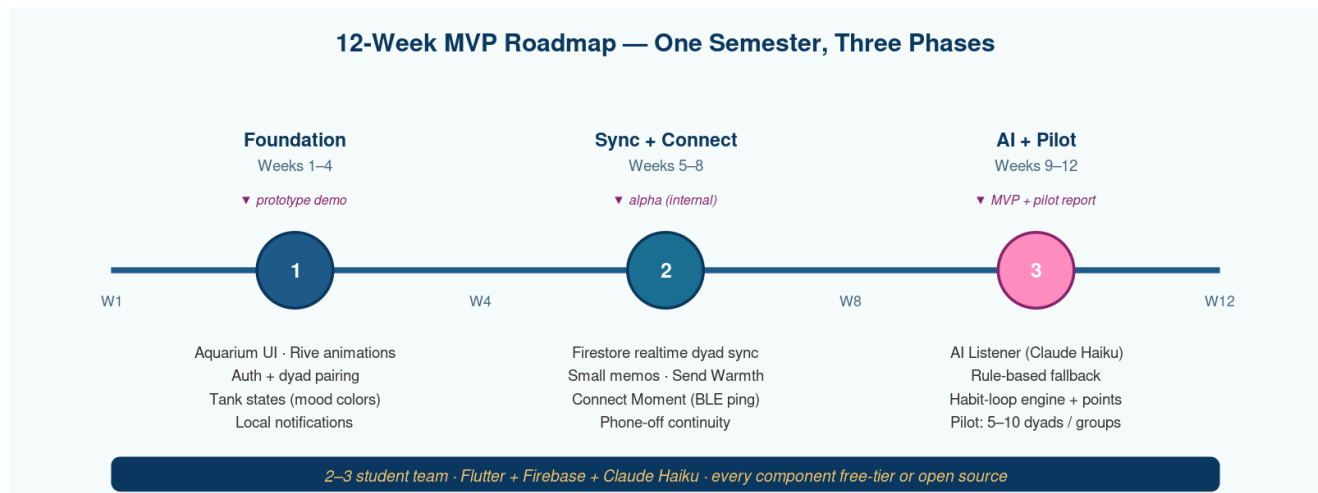


Figure 3. System architecture — free-tier components, on-device privacy, graceful AI degradation.

- **Frontend.** Flutter (single codebase, iOS + Android) with Rive for the state-driven, lightweight aquarium animation; *flutter_local_notifications* for low-pressure nudges; Bluetooth Low Energy for the Connect Moment proximity ping.
- **Backend.** Firebase — Firestore gives realtime multi-device tank sync out of the box; Cloud Functions run the habit-loop engine and points ledger; FCM handles push. Generous free tier. (Alternative: Supabase for SQL portability.)
- **AI layer.** Gemini for nudge generation — short and infrequent, so cost is centavos per user per month; prompt-engineered with dyad context (love language, cadence, mood), no fine-tuning needed. A **rule-based fallback** ("no interaction in 4 days → suggest memo") keeps the app fully functional without any LLM call.
- **Privacy-first signals.** Apple ScreenTime / Android UsageStatsManager processed on-device; in-app cadence only (never SMS/Messenger content); HealthKit / Google Fit for activity milestones — "hike completed," never raw GPS. BLE proximity peer-to-peer; no location leaves the phone.
- **Vouchers (Phase 2).** Start manual — 2–3 partner cafés with unique promo codes in Firestore; no payment integration needed for MVP.

Timeline. A 2–3 person student team ships a functional MVP in one semester — Weeks 1–4 core app + aquarium UI; Weeks 5–8 dyad sync, memos, tank states, BLE Connect Moment; Weeks 9–12 AI listener, habit loop, and a pilot with 5–10 real dyads/small groups. Every component is free-tier or open source; the hardest technical problem (realtime sync + animation + peer-to-peer BLE) is already solved by Firebase + Rive + platform APIs.



Design Principles & Success Metrics

- **Autonomy-supportive, never controlling** — engineered for the research-backed sweet spot of high awareness with high autonomy support.
- **Low vulnerability cost** — every interaction has a one-tap, wordless version; nobody is forced to explain themselves.
- **No guilt mechanics** — streaks never shame, the fish never dies, absence is never punished; KIBO must not digitally recreate obligation pressure.
- **Rewards real presence** — the biggest payoff (Connect Moment nutrients + capsules) only unlocks when users are actually together, phones off.

We measure success by **relationship outcomes, not engagement**: interaction reciprocity (not raw volume); self-reported closeness and reduced communication anxiety (pre/post pilot surveys); weekly active dyads/groups retaining the habit loop; Connect Moment frequency; and — our favorite metric — **reduction in surveillance-tool usage and in-app screen time** within groups. KIBO succeeds when people need less technology between them, not more. *Disconnected, by design.*

Key References

- [1] World Health Organization. Commission on Social Connection: 1 in 6 people globally affected by loneliness; ~100 deaths per hour linked to social disconnection (2024–2026).
- [2] Mind Health Report (2026): 72% of Filipino youth report loneliness/sadness despite ~7.5h/day screen time. BusinessWorld PH coverage, June 2026.
- [3] Dwyer, Kushlev & Dunn. "Smartphone use undermines enjoyment of face-to-face social interactions." *Journal of Experimental Social Psychology* (2018).
- [4] Roberts & David. "My life has become a major distraction from my cell phone" — Phubbing and relationship satisfaction. *Computers in Human Behavior*; UConn / Institute for Family Studies analyses (2016–2026).
- [5] Britannica overview of friendship formation research (Berscheid, Rawlins) — convenience as the dominant predictor of interaction frequency; convenience-based friendships rarely deepen.
- [6] Cox, Survey Center on American Life / American Perspectives Survey — decline of close friendships in the U.S. from 1990 baseline; "friendship recession" analyses (Harvard Kennedy School Leadership & Happiness Lab, 2025).
- [7] Uhls et al. "Five days at outdoor education camp without screens improves preteen skills with nonverbal emotion cues." *Computers in Human Behavior* (2014); follow-up research on empathy and screen time.
- [8] Pew Research Center (2022): 65% of Gen Z say they had to re-learn social skills post-pandemic.
- [9] Weiser & Brown. "The Coming Age of Calm Technology." Xerox PARC (1996) — informs without demanding attention.
- [10] IDC / industry ambient computing forecasts through 2026: 60% of developed-market households with ambient tech; ambient market ~US\$85B in 2026 (26% CAGR).
- [11] Philippine Statistics Authority — Survey on Overseas Filipinos 2024: 2.19M OFWs, +1.5% YoY.
- [12] Finch (finchcare.com) — virtual-pet self-care app, ~US\$30M ARR, bootstrapped; freemium at US\$9.99/mo; validating habit-loop virtual-pet monetization.
- [13] Tandfonline / PMC (2024–2025). Digital device engagement, digital stress, and coping strategies among Filipino college students (N=384); social anxiety, smartphone addiction, and loneliness in college populations.